

Neutron-Gamma Discrimination Using Two Complementary Methods

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Abstract

^{252}Cf decays in spontaneous fission with emission of neutrons and gamma rays. Our goal was to discriminate these two kinds of neutral particles thanks to two methods. One is the fission-neutron time of flight (ToF) by using scintillators which consider the prompt γ rays as a fast timing start signal and neutrons as a slow timing stop signal. The second method uses the fact that neutrons and γ rays excite different states of organic scintillators, the fluorescence and phosphorescence decays respectively. Thus, the shape of the signal differs: this is the Pulse Shape Discrimination (PSD). By applying these two complementary methods and comparing their results, we were able to discriminate neutrons and γ rays.

1 Introduction

^{252}Cf decays through two channels: α decay (96.9 % of cases) and spontaneous fission (SF) (3.1 % of cases) [1]. Neutrons and γ rays are emitted only in the latter case, with on average about 3.7 neutrons and 9 prompt γ rays per fission [2]. ^{252}Cf has a half-life of $T_{1/2} = 2.647$ years [1]. The activity of our source was $A = 362 \pm 4$ kBq, the uncertainty accounting for the decay of the source over the one-month duration of the experiment. Uncertainties on the half-life and on the initial activity are considered negligible.

The time-of-flight (ToF) method requires two detectors: one for γ rays, providing the start signal of the acquisition, and one for neutrons, providing the stop signal. The delay between the two signals defines the time of flight. γ rays interact through the electromagnetic interaction, whereas neutrons interact via the strong interaction. γ -ray detection is therefore best achieved with an inorganic scintillator featuring a high atomic number Z and a fast fluorescence decay component. A BaF_2 scintillator was thus chosen for its mean atomic number $Z = 46$. A high mean atomic number is indeed essential to favor the photoelectric effect, whose cross section scales as $\sigma_{pe} \propto Z^5$. For neutron detection, a highly hydrogenated organic scintillator is preferable,

as it maximizes the momentum transfer between the neutron and the proton. An NE213 liquid organic scintillator was therefore used.

The time-of-flight measurement relies on the fact that γ rays travel at the speed of light c , whereas neutrons travel at a lower velocity. The measured time is therefore the delay between the detection of the two signals. Once the time of flight Δt is known, the neutron kinetic energy can be derived:

$$E_k = \frac{1}{2}m_n v^2 \text{ with } v = \frac{L}{\Delta t}, \text{ hence } E_k = \frac{1}{2}m_n \left(\frac{L}{\Delta t} \right)^2 \quad (1)$$

where L is the distance between the source and the organic scintillator, and $m_n = 939.6 \text{ MeV}/c^2$ is the neutron mass. Equation (1) holds in the non-relativistic regime. Indeed, the mean neutron energy is $\langle E_n \rangle \approx 2.14 \text{ MeV}$ [2], so that $\gamma - 1 \approx 2.3 \times 10^{-3} \ll 1$, which corresponds to neutrons travelling at about 6.6 % of the speed of light.

Pulse Shape Discrimination (PSD) relies on the fact that γ rays and neutrons do not excite the same states of the organic scintillator molecules. γ rays excite the singlet states of the organic molecules, which decay by fluorescence with a short time constant ($\tau = 3 \text{ ns}$), whereas neutrons excite the triplet states, which decay more slowly through phosphorescence ($\tau = 270 \text{ ns}$) [3]. Neutron-induced pulses therefore exhibit a longer tail, and hence a higher ratio of tail area to total area. Neutrons and γ rays can then be discriminated by plotting this area ratio as a function of the total area [4]. This method requires only a single detector, which must be able to detect both neutral particles: the NE213 liquid organic scintillator.

2 Experimental setup

The experimental setup is shown in Fig. 1.

The operating range of both detectors was determined using atmospheric muon coincidences. The high voltage was set to 2100 V for the inorganic scintillator and to 1100 V for the organic one. The distance L_2 was chosen as large as possible in order to minimize the

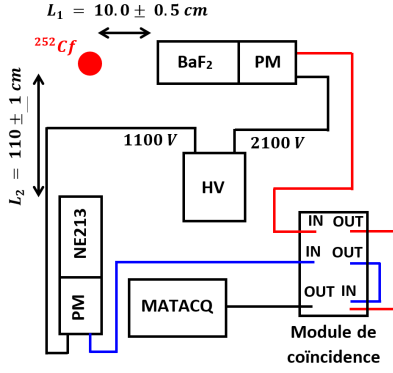


Figure 1: Schematic of the experimental setup

relative uncertainty on the neutron time-of-flight measurement, and thus improve the precision of the neutron energy estimate. Two channels of the coincidence module are used to generate logic gates with a width of 150 ns. These gates must indeed be wider than both the mean neutron time of flight (≈ 50 ns) and the PMT response time (≈ 30 ns), while remaining narrower than the mean interval between two successive SF events ($\approx 90 \mu\text{s}$). A ^{60}Co source, which decays by emitting two γ rays less than 1 ps apart [5], was used to align the coincidence signals by adjusting the cable lengths. This removes any time offset between the two detectors. The signals from both detectors and from the coincidence module are acquired with the MATAcq software. The post-trigger is set so as to record 50 ns of signal before the peak and about $1 \mu\text{s}$ after it. The signal can thus be integrated over more than three times the characteristic phosphorescence time, while keeping a sampling step of 0.5 ns, which provides a suitable time-of-flight resolution.

The ^{60}Co source was also used to perform an energy calibration of the NE213 detector, by identifying the Compton edge and the backscatter peak in the acquired spectrum.

3 Results

3.1 Time of flight

The time-of-flight results are presented in Fig. 2 and correspond to 45 000 acquisitions. The red region corresponds to $\gamma\gamma$ coincidences. A Gaussian fit was performed on the peak attributed to γ rays: it is centered at 21 ± 1 ns with a standard deviation of 2.8 ± 0.4 ns. The green region corresponds to events that cannot be identified by time of flight. It is centered on the minimum between the two distributions and spans 5 times the standard deviation of the γ distribution. The broader blue band, extending from 50 to 150 ns, corresponds to neutrons. This second peak is much wider because

the neutron energy distribution ranges from about a hundred keV to a few MeV. The neutron energy distribution

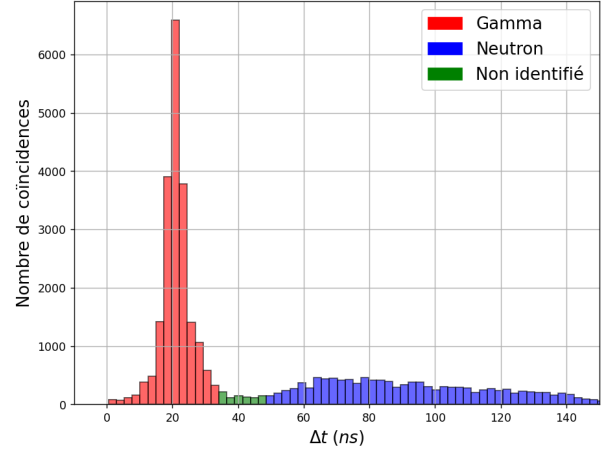


Figure 2: Time-of-flight histogram

the neutron energy distribution can be derived from the neutron ToF measurements using Eq. (1). For spontaneous fission into two fragments of different masses, the neutron energy distribution follows a Watt spectrum [6]:

$$N(E_n) \propto \exp(-E_n/T) \sinh(\sqrt{b E_n}) \quad (2)$$

where T is the nuclear temperature and b a spectrum-dependent parameter. However, this distribution is distorted by the efficiency of the organic scintillator, which depends on the neutron energy:

$$N_{\text{meas}}(E_n) = N(E_n) \times \text{Eff}(E_n) \quad (3)$$

where $\text{Eff}(E_n)$ is the detector efficiency. To determine it, the measured neutron energy distribution was normalized to its integral and then divided by the Watt distribution [7]. The resulting detector efficiency curve is shown in Fig. 3. This curve appears consistent with typical efficiency curves of organic scintillators [6], indicating that the detected neutrons are compatible with a Watt distribution.

3.2 PSD

Three integration windows are used in this method: TOT, DELAYED and FAST. The TOT window integrates the whole signal. The DELAYED window integrates from the time at which the signal falls below 20% of its maximum amplitude on the trailing edge to the end of the signal. The FAST window integrates from the time at which the signal reaches 20% of its maximum amplitude on the rising edge to the time of the signal maximum.

Fig. 4 shows the DELAYED/TOT ratio as a function of TOT. A Figure of Merit (FOM) can be computed to

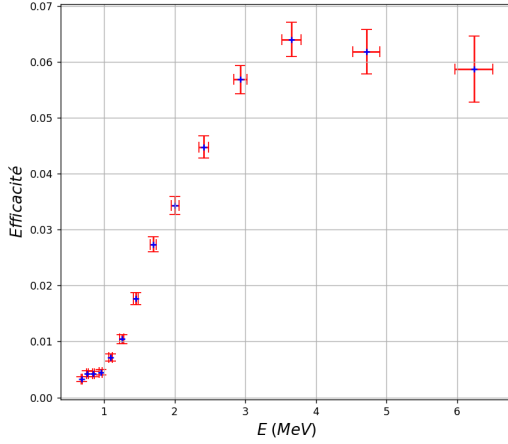


Figure 3: Neutron detection efficiency of the NE213 scintillator

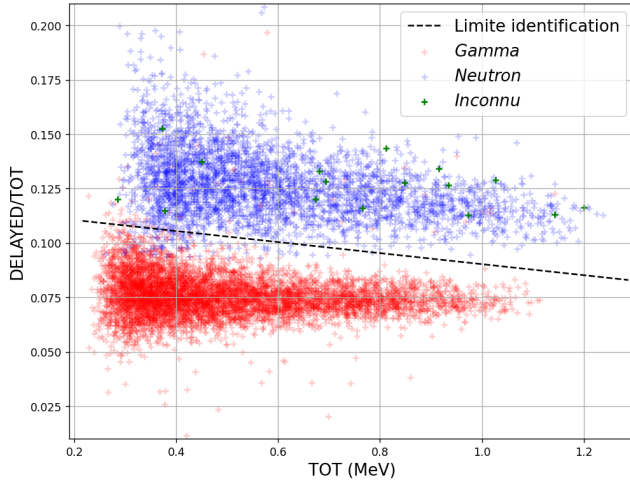


Figure 4: DELAYED/TOT as a function of TOT

quantify the separation between neutrons and γ rays:

$$\text{FOM} = \frac{|\mu_\gamma - \mu_n|}{\sigma_\gamma + \sigma_n} \quad (4)$$

where μ_γ (resp. μ_n) is the mean of the γ -ray (resp. neutron) distribution, and σ_γ (resp. σ_n) is the standard deviation of the γ -ray (resp. neutron) distribution. This FOM has a clear physical meaning. When it equals unity, the two distributions overlap and misidentifications occur. A FOM greater than 2 can be considered necessary to reasonably conclude on the identity of a particle. The FOM is computed by performing Gaussian fits of both distributions in the 1D histogram of the DELAYED/TOT ratio, as shown in Fig. 5. The FOM was evaluated for various integration gates, and the optimal DELAYED window was found to extend from 33.5 ns to 360 ns after the signal falls below 20 % of its maximum. Under these conditions, we obtain $\text{FOM} = 2.26 \pm 0.01$. The FAST/TOT representation

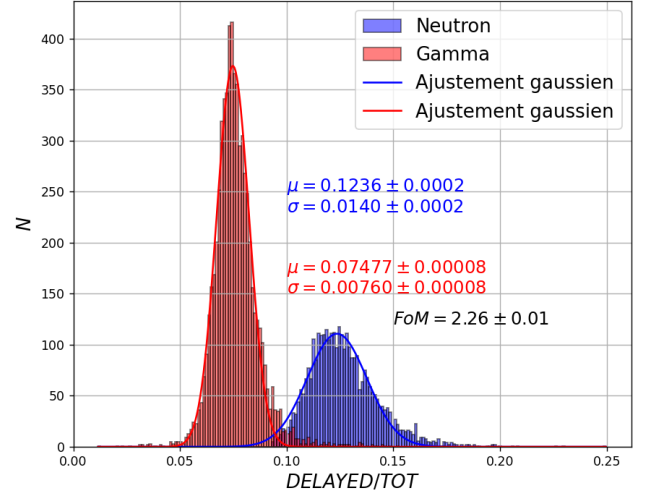


Figure 5: Histogram of the DELAYED/TOT ratio

as a function of TOT yields at best $\text{FOM} = 1.94 \pm 0.02$, and therefore does not improve the discrimination between neutrons and γ rays.

Some undesirable acquisitions were also removed by plotting DELAYED/TOT as a function of the time of flight, for events with a time of flight greater than 34 ns (corresponding to the lower limit of the green region in Fig. 2) and a DELAYED/TOT ratio typical of γ rays. Indeed, some events identified as neutrons by the ToF method are identified as γ rays by PSD. These are starts or stops triggered by stray photons, which produce a detection background present at all times of flight and clearly distinguishable in this representation. These events were therefore discarded.

3.3 Consistency of the two methods

Table 1 presents the level of agreement between the two methods. PSD is applied over the energy ranges given in the first column of Table 1. In Fig. 4, an event is classified as a neutron if it lies above the boundary between the two distributions (black line in Fig. 4), and as a γ ray otherwise. Since each event has already been identified by the ToF method, the percentages given are the ratio of the number of events identified as neutrons (resp. γ rays) by PSD to the number of events identified as neutrons (resp. γ rays) by ToF. PSD therefore becomes increasingly reliable at higher energies. An energy of at least 0.63 MeV is required to reach an agreement above 99 % for both γ -ray and neutron identification.

Energy (MeVee)	Neutron (%)	Gamma (%)
$0.28 \pm 0.02 - 0.45 \pm 0.02$	93.9 ± 0.8	98.6 ± 0.2
$0.45 \pm 0.02 - 0.63 \pm 0.02$	97.7 ± 0.5	98.8 ± 0.3
$0.63 \pm 0.02 - 0.80 \pm 0.03$	99.2 ± 0.4	99.2 ± 0.3
$0.80 \pm 0.03 - 1.32 \pm 0.04$	99.6 ± 0.2	99.4 ± 0.3

Table 1: Agreement between the two identification methods

4 Discussion

4.1 Time of flight

The γ -ray peak is centered at 21 ± 1 ns, and the maximum of the neutron peak lies at 71 ± 3 ns. Theoretically, however, the γ -ray peak is expected at 3.33 ± 0.05 ns given the distance between the source and the two detectors (the uncertainty arising from the parameters L_1 and L_2), and the neutron peak at 54 ± 1 ns [2]. The expected separation between the two peaks is 51 ± 1 ns, and the measured one is 50 ± 4 ns: the two values are in agreement. We therefore have a systematic offset of 17 ± 4 ns. However, the cable lengths of the two detectors had been adjusted so as not to introduce any delay between them. This systematic offset is most likely due to the coincidence module. Since the width of the logic gates could not be adjusted with the discriminator, two channels of the coincidence module were used to generate them. Switching modules most likely introduced this systematic delay between the two signals.

Using the ^{60}Co energy calibration, we determined that 80 % of the detected γ rays from ^{252}Cf have an energy below 0.67 ± 0.02 MeV, in good agreement with [6].

The efficiency of the organic detector, computed from the neutron energy distribution (see Fig. 3), is consistent with the detector specifications (an efficiency slightly above 6 % at 3 MeV). Moreover, 3478 ± 59 neutrons were detected over 40 ± 1 minutes, whereas the expected value is $(4.9 \pm 1.1) \times 10^3$ for a mean detector efficiency of 4.8 ± 0.6 %. This yields a normalized deviation $\text{N.D.} = (N_{n,th} - N_{n,exp}) / \sqrt{U^2(N_{n,th}) + U^2(N_{n,exp})} \approx 1.3$. The two values therefore agree within two standard deviations. Hence, the consistency of both the detector efficiency and the number of detected neutrons, once events in the overlap region of the two distributions have been excluded (green region in Fig. 2), shows that the time-of-flight method reliably identifies neutrons. This holds in particular for lower-energy neutrons (below 5 MeV), whose longer times of flight (above 34 ns) allow a clear separation from γ rays. For higher-energy neutrons, PSD is more effective.

However, a few events with very short times of flight (< 17 ns) are observed, which cannot be identified as γ rays. Just as background sometimes triggers the stop in

the NE213, some stray photons trigger the start in the BaF_2 shortly before a neutron crosses the organic scintillator. Such events occur regularly because the BaF_2 is rather noisy, and they are spread uniformly over all time-of-flight values. They were also partly removed by the cut described at the end of Sec. 3.2.

4.2 PSD

First, Table 1 clearly shows that PSD is more reliable at high energy than at low energy. The two methods agree in more than 99 % of cases above 0.63 ± 0.02 MeVee. Events that could not be identified by the ToF method were identified by PSD. Indeed, the green points in Fig. 2 are all identified as neutrons by PSD. Their times of flight range from 23 ± 4 ns to 28 ± 4 ns, corresponding to energies from 8 ± 3 MeV to 12 ± 5 MeV. These are indeed high-energy neutrons, which could not be identified by ToF but were successfully identified by PSD.

5 Conclusion and outlook

While PSD provides a clear discrimination between fast neutrons and γ rays at high energy, the lower-energy region is more challenging. Time of flight proves to be a very effective method in this regime, provided that coincidences caused by background in both detectors have first been removed using a DELAYED/TOT versus ToF diagram. A possible extension of this work would be the implementation of a neural network able to automatically discriminate between neutrons and γ rays, trained through repeated sorting and comparison of the acquisitions.

References

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